Some definitions

set is finite sets implemented as Lists.

$$
\begin{aligned}
\mathit{Subst} &:= \mathit{set}\ (\mathit{Var} * \mathit{exp}) \\
\mathit{Equation} &:= \mathit{set}\ ((\mathit{exp} * \mathit{exp}) + (\mathit{sexp} * \mathit{sexp})) \\
\mathit{Problem} &:= \mathit{set}\ \mathit{Equation} \\
\mathit{Tuple} &:= \mathit{Subst} * \mathit{Problem} \\
\_\langle\_\rangle &: \mathit{exp} \to \mathit{Subst} \to \mathit{exp} \\
\_\langle\_\rangle &: \mathit{sexp} \to \mathit{Subst} \to \mathit{sexp} \\
\_\langle\_\rangle &: \mathit{Problem} \to \mathit{Subst} \to \mathit{Problem} \\
\_\ @\ \_ &: \mathit{Subst} \to \mathit{Var} \to \mathit{exp} \\
\mathit{dom} &: \mathit{Subst} \to \mathit{set}\ \mathit{Var} \\
\mathit{smatch} &: \mathit{Tuple} \to \mathit{Tuple} \to \mathit{Prop}
\end{aligned}
$$

smatch steps are classified into three

smatch steps are classified into three

Congruence steps

$$\begin{aligned} \mathit{smatch\_app\_congr} : \ & (s \ t =_? s' \ t') \in P \to \\ & \mathit{smatch} \ (S, P) \ (S, P + (s =_? s') + (t =_? t') \setminus (s \ t =_? s' \ t')) \end{aligned}$$

smatch steps are classified into three

Congruence steps

$$\begin{aligned} \mathit{smatch\_app\_congr} : \ & (s\ t =_? s'\ t') \in P \rightarrow \\ & \mathit{smatch}\ (S, P)\ (S, P + (s =_? s') + (t =_? t') \setminus (s\ t =_? s'\ t')) \end{aligned}$$

σ-min steps

$$\begin{aligned} \mathit{smatch\_subst\_comp} : \ & (s[\sigma \circ \tau] =_? s'[\rho]) \in P \rightarrow \\ & \mathit{smatch}\ (S, P)\ (S, P + (s[\sigma][\tau] =_? s'[\rho]) \setminus (s[\sigma \circ \tau] =_? s'[\rho])) \end{aligned}$$

smatch steps are classified into three

Congruence steps

$$\begin{aligned} \textit{smatch\_app\_congr}:\ & (s\ t =_? s'\ t') \in P \rightarrow \\ & \textit{smatch}\ (S, P)\ (S, P + (s =_? s') + (t =_? t')\ \setminus\ (s\ t =_? s'\ t')) \end{aligned}$$

σ-min steps

$$\begin{aligned} \textit{smatch\_subst\_comp}:\ & (s[\sigma \circ \tau] =_? s'[\rho]) \in P \rightarrow \\ & \textit{smatch}\ (S, P)\ (S, P + (s[\sigma][\tau] =_? s'[\rho]) \setminus (s[\sigma \circ \tau] =_? s'[\rho])) \end{aligned}$$

Instantiation case

$$\begin{aligned} \textit{smatch\_inst}:\ & (s =_? X) \in P \rightarrow \\ & X \notin \textit{lhvars\_of}\ \ P \rightarrow \\ & S' =_{\sigma_{min}} S \gg \{(X, s)\} \rightarrow \\ & \textit{smatch}\ (S, P)\ (S', (P \setminus (s =_? X))\langle\{(X, s)\}\rangle) \end{aligned}$$

Preservation

Validity

$$valid\_tuple\ (S, P) := \underline{(dom\ S) \cap (vars\_of\ \ P) = \phi}$$

Matching solution

$$match\_sol\ \ Sl\ (S, P)\ \ := \underline{(s =_? t) \in P \to s =_{\sigma_{min}} t\langle Sl\rangle}\ \wedge$$
$$\underline{\exists S', S \gg S' =_{\sigma_{min}} Sl}$$

Preservation theorem

$$\begin{aligned} match\_pres\ \ :\ & valid\_tuple\ \ T \to \\ & (dom\ \ Sl) \cap (lhvars\_of\ \ P) \to \\ & smatch\ \ T\ \ T' \to \\ & match\_sol\ \ Sl\ \ T' \to \\ & match\_sol\ \ Sl\ \ T \end{aligned}$$

Soundness

smatch is sound

$$\begin{aligned} match\_pres \;:\; & valid\_tuple\ T \rightarrow \\ & (dom\ Sl) \cap (lhvars\_of\ P) \rightarrow \\ & star\ smatch\ T\ T' \rightarrow \\ & match\_sol\ Sl\ T' \rightarrow \\ & match\_sol\ Sl\ T \end{aligned}$$

Proof proceeds by case analysis on *smatch*

Inst case

$$
\begin{array}{l}
H_1 \;:\; (dom\ S)\ \cap\ vars\_of\ \ P = \{\} \\
H_2 \;:\; (dom\ Sl) \cap\ lhvars\_of\ \ P = \{\} \\
H_3 \;:\; X \notin lhvars\_of\ \ P \\
H_4 \;:\; match\_sol\ Sl\ (S \gg \{(X, s)\},\ (P \setminus (s =_? X))\langle\{(X, s)\}\rangle \\
\hline
match\_sol\ Sl\ \ P
\end{array}
$$

Inst case (Unfolded *match_sol*)

$$
\begin{array}{l}
H_1 \ : \ (dom\ S) \ \cap \ vars\_of\ \ P = \{\} \\
H_2 \ : \ (dom\ Sl) \cap \ lhvars\_of\ \ P = \{\} \\
H_3 \ : \ X \notin lhvars\_of\ \ P \\
H_4 \ : \ \forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \to s =_{\sigma_{min}} t\langle Sl\rangle \\
H_5 \ : \ \exists S', Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
\text{-----------------------------------} \\
\forall s_0, t_0, (s_0 =_? t_0) \in P \to s_0 =_{\sigma_{min}} t_0\langle Sl\rangle \quad \wedge \\
\exists S', Sl =_{\sigma_{min}} S \gg S'
\end{array}
$$

Inst case (Unfolded *match_sol*)

$$
\begin{array}{l}
H_1 \ : \ (dom\ S) \ \cap \ vars\_of\ \ P = \{\} \\
H_2 \ : \ (dom\ Sl) \cap \ lhvars\_of\ \ P = \{\} \\
H_3 \ : \ X \notin lhvars\_of\ \ P \\
H_4 \ : \ \forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \to s =_{\sigma_{min}} t\langle Sl\rangle \\
H_5 \ : \ \exists S', Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
------------------------------------ \\
\forall s_0, t_0, (s_0 =_? t_0) \in P \to s_0 =_{\sigma_{min}} t_0\langle Sl\rangle \quad \wedge \\
\exists S', Sl =_{\sigma_{min}} S \gg S'
\end{array}
$$

Second conjunct is solvable associativity property of substitution composition.

Inst case continued..

$$
\begin{array}{l}
H_1 \ : \ (dom\ S) \ \cap \ vars\_of \ \ P = \{\} \\
H_2 \ : \ (dom\ Sl) \cap \ lhvars\_of \ \ P = \{\} \\
H_3 \ : \ X \notin lhvars\_of \ \ P \\
H_4 \ : \ \forall\ s' \ t', (s' =_? t') \in (P \backslash (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{min}} t' \langle Sl \rangle \\
H_5 \ : \ Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
s_0, t_0 \ : \ \texttt{exp} \\
H_6 \ : \ (s_0 =_? t_0) \in P \\
\hline
s_0 =_{\sigma_{min}} t_0 \langle Sl \rangle
\end{array}
$$

Inst case continued..

$$
\begin{array}{l}
H_1 \ : \ (dom\ S) \ \cap \ vars\_of \ \ P = \{\} \\
H_2 \ : \ (dom\ Sl) \cap \ lhvars\_of \ \ P = \{\} \\
H_3 \ : \ X \notin lhvars\_of \ \ P \\
H_4 \ : \ \forall\ s' \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X,s)\}\rangle \rightarrow s' =_{\sigma_{min}} t'\langle Sl\rangle \\
H_5 \ : \ Sl =_{\sigma_{min}} (S \gg \{(X,s)\}) \gg S' \\
s_0, t_0 \ : \ \texttt{exp} \\
H_6 \ : \ (s_0 =_? t_0) \in P \\
\text{-----------------------------------} \\
s_0 =_{\sigma_{min}} t_0\langle Sl\rangle
\end{array}
$$

Case analysis : $\{s_0 =_? t_0 \ = \ s =_? X\} + \{s_0 =_? t_0 \ \neq s =_? X\}$

Case 1: $s_0 =_? t_0 \;=\; s =_? X$

$H_1 \;:\; (dom\ S) \;\cap\; vars\_of\ \ P = \{\}$
$H_2 \;:\; (dom\ Sl) \cap\; lhvars\_of\ \ P = \{\}$
$H_3 \;:\; X \notin lhvars\_of\ \ P$
$H_4 \;:\; \forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X,s)\}\rangle \to s' =_{\sigma_{min}} t'\langle Sl\rangle$
$H_5 \;:\; Sl =_{\sigma_{min}} (S \gg \{(X,s)\}) \gg S'$
$s_0, t_0 \;:\; \texttt{exp}$
$H_6 \;:\; (s =_? X) \in P$
———————————————————————————————————
$s =_{\sigma_{min}} X\langle Sl\rangle$

Case 1: $s_0 =_? t_0 \;=\; s =_? X$

$$
\begin{array}{l}
H_1 \;:\; (dom\ S)\ \cap\ vars\_of\ \ P = \{\} \\
H_2 \;:\; (dom\ Sl) \cap\ lhvars\_of\ \ P = \{\} \\
H_3 \;:\; X \notin lhvars\_of\ \ P \\
H_4 \;:\; \forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{min}} t' \langle Sl \rangle \\
H_5 \;:\; Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
s_0, t_0 \;:\; \texttt{exp} \\
H_6 \;:\; (s =_? X) \in P \\
\text{-----------------------------------} \\
s =_{\sigma_{min}} X \langle Sl \rangle
\end{array}
$$

$s =_{\sigma_{min}} Sl@X$ is not immediate

Case 1: $s_0 =_? t_0 \;=\; s =_? X$

H_1 : $(dom\ S) \cap vars\_of\ P = \{\}$
H_2 : $(dom\ Sl) \cap lhvars\_of\ P = \{\}$
H_3 : $X \notin lhvars\_of\ P$
H_4 : $\forall\ s'\ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \rightarrow s' =_{\sigma_{min}} t'\langle Sl\rangle$
H_5 : $Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S'$
s_0, t_0 : `exp`
H_6 : $(s =_? X) \in P$
———————————————————————————————————
$s =_{\sigma_{min}} X\langle Sl\rangle$

$s =_{\sigma_{min}} Sl@X$ is not immediate

1. $Y \in (vars\_of\ s) \rightarrow Y \notin dom\ Sl$, from H_2 and H_6

Case 1: $s_0 =_? t_0 \ = \ s =_? X$

H_1 : $(dom\ S)\ \cap\ vars\_of\ \ P = \{\}$
H_2 : $(dom\ Sl) \cap\ lhvars\_of\ \ P = \{\}$
H_3 : $X \notin lhvars\_of\ \ P$
H_4 : $\forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \rightarrow s' =_{\sigma_{min}} t'\langle Sl\rangle$
H_5 : $Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S'$
s_0, t_0 : $\mathtt{exp}$
H_6 : $(s =_? X) \in P$
——————————————————————————————
$s =_{\sigma_{min}} X\langle Sl\rangle$

$s =_{\sigma_{min}} Sl@X$ is not immediate

1. $Y \in (vars\_of\ \ s) \rightarrow Y \notin dom\ Sl$, from H_2 and H_6
2. $Y \in (vars\_of\ \ s) \rightarrow Y \notin dom\ (S \gg \{(X, s)\})$, from H_1 and H_3.

Case 1: $s_0 =_? t_0 \; = \; s =_? X$

H_1 : $(dom\ S)\ \cap\ vars\_of\ \ P = \{\}$
H_2 : $(dom\ Sl) \cap\ lhvars\_of\ \ P = \{\}$
H_3 : $X \notin lhvars\_of\ \ P$
H_4 : $\forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \rightarrow s' =_{\sigma_{min}} t'\langle Sl\rangle$
H_5 : $Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S'$
s_0, t_0 : `exp`
H_6 : $(s =_? X) \in P$
——————————————————————————————
$s =_{\sigma_{min}} X\langle Sl\rangle$

$s =_{\sigma_{min}} Sl@X$ is not immediate

1. $Y \in (vars\_of\ \ s) \rightarrow Y \notin dom\ Sl$, from H_2 and H_6
2. $Y \in (vars\_of\ \ s) \rightarrow Y \notin dom\ (S \gg \{(X, s)\})$, from H_1 and H_3.
3. $(S \gg \{(X, s)\})@X = s$, from H_1 and H_6

Case 1: $s_0 =_? t_0 \ = \ s =_? X$

$$
\begin{array}{l}
H_1 \ : \ (dom\ S) \ \cap \ vars\_of \ \ P = \{\} \\
H_2 \ : \ (dom\ Sl) \cap \ lhvars\_of \ \ P = \{\} \\
H_3 \ : \ X \notin lhvars\_of \ \ P \\
H_4 \ : \ \forall\ s' \ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \to s' =_{\sigma_{min}} t'\langle Sl\rangle \\
H_5 \ : \ Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
s_0, t_0 \ : \ \texttt{exp} \\
H_6 \ : \ (s =_? X) \in P \\
\text{------------------------------------} \\
s =_{\sigma_{min}} X\langle Sl\rangle
\end{array}
$$

$s =_{\sigma_{min}} Sl@X$ is not immediate

1. $Y \in (vars\_of \ \ s) \to Y \notin dom\ Sl$, from H_2 and H_6
2. $Y \in (vars\_of \ \ s) \to Y \notin dom\ (S \gg \{(X, s)\})$, from H_1 and H_3.
3. $(S \gg \{(X, s)\})@X = s$, from H_1 and H_6
4. $((S \gg \{(X, s)\}) \gg S')@X = ((S \gg \{(X, s)\})@X)\langle S'\rangle = s\langle S'\rangle$, from 3 and lookup property

Case 1: $s_0 =_? t_0 \;=\; s =_? X$

$$
\begin{array}{l}
H_1 \;:\; (dom\ S) \,\cap\, vars\_of\ \ P = \{\} \\
H_2 \;:\; (dom\ Sl) \cap\ lhvars\_of\ \ P = \{\} \\
H_3 \;:\; X \notin lhvars\_of\ \ P \\
H_4 \;:\; \forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \rightarrow s' =_{\sigma_{min}} t'\langle Sl\rangle \\
H_5 \;:\; Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
s_0, t_0 \;:\; \texttt{exp} \\
H_6 \;:\; (s =_? X) \in P \\
\text{------------------------------------} \\
s =_{\sigma_{min}} X\langle Sl\rangle
\end{array}
$$

$s =_{\sigma_{min}} Sl@X$ is not immediate

1. $Y \in (vars\_of\ \ s) \rightarrow Y \notin dom\ Sl$, from H_2 and H_6
2. $Y \in (vars\_of\ \ s) \rightarrow Y \notin dom\ (S \gg \{(X, s)\})$, from H_1 and H_3.
3. $(S \gg \{(X, s)\})@X = s$, from H_1 and H_6
4. $((S \gg \{(X, s)\}) \gg S')@X = ((S \gg \{(X, s)\})@X)\langle S'\rangle = s\langle S'\rangle$, from 3 and lookup property
5. $Y \in (vars\_of\ \ s) \rightarrow S'@Y =_{\sigma_{min}} Y$, from a substitution property

Case 1: $s_0 =_? t_0 \; = \; s =_? X$

$$
\begin{array}{l}
H_1 \; : \; (dom \; S) \; \cap \; vars\_of \; \; P = \{\} \\
H_2 \; : \; (dom \; Sl) \cap \; lhvars\_of \; \; P = \{\} \\
H_3 \; : \; X \notin lhvars\_of \; \; P \\
H_4 \; : \; \forall \; s' \; \; t', (s' =_? t') \in (P \backslash (s =_? X)) \langle \{(X, s)\} \rangle \rightarrow s' =_{\sigma_{min}} t' \langle Sl \rangle \\
H_5 \; : \; Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
s_0, t_0 \; : \; \texttt{exp} \\
H_6 \; : \; (s =_? X) \in P \\
\text{------------------------------------} \\
s =_{\sigma_{min}} X \langle Sl \rangle
\end{array}
$$

$s =_{\sigma_{min}} Sl@X$ is not immediate

1. $Y \in (vars\_of \;\; s) \rightarrow Y \notin dom \; Sl$, from H_2 and H_6
2. $Y \in (vars\_of \;\; s) \rightarrow Y \notin dom \; (S \gg \{(X, s)\})$, from H_1 and H_3.
3. $(S \gg \{(X, s)\})@X = s$, from H_1 and H_6
4. $((S \gg \{(X, s)\}) \gg S')@X = ((S \gg \{(X, s)\})@X)\langle S' \rangle = s\langle S' \rangle$, from 3 and lookup property
5. $Y \in (vars\_of \;\; s) \rightarrow S'@Y =_{\sigma_{min}} Y$, from a substitution property
6. $s =_{\sigma_{min}} s\langle S' \rangle$, from 5

Case 1: $s_0 =_? t_0 \;=\; s =_? X$

$$
\begin{array}{l}
H_1 \;:\; (dom\ S) \;\cap\; vars\_of\ \ P = \{\} \\
H_2 \;:\; (dom\ Sl) \cap\; lhvars\_of\ \ P = \{\} \\
H_3 \;:\; X \notin lhvars\_of\ \ P \\
H_4 \;:\; \forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \to s' =_{\sigma_{min}} t'\langle Sl\rangle \\
H_5 \;:\; Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
s_0, t_0 \;:\; \mathtt{exp} \\
H_6 \;:\; (s =_? X) \in P \\
\text{------------------------------------} \\
s =_{\sigma_{min}} X\langle Sl\rangle
\end{array}
$$

$s =_{\sigma_{min}} Sl@X$ is not immediate

1. $Y \in (vars\_of\ \ s) \to Y \notin dom\ Sl$, from H_2 and H_6
2. $Y \in (vars\_of\ \ s) \to Y \notin dom\ (S \gg \{(X, s)\})$, from H_1 and H_3.
3. $(S \gg \{(X, s)\})@X = s$, from H_1 and H_6
4. $((S \gg \{(X, s)\}) \gg S')@X = ((S \gg \{(X, s)\})@X)\langle S'\rangle = s\langle S'\rangle$, from 3 and lookup property
5. $Y \in (vars\_of\ \ s) \to S'@Y =_{\sigma_{min}} Y$, from a substitution property
6. $s =_{\sigma_{min}} s\langle S'\rangle$, from 5

Property mentioned in 5:
$\forall\ X\ S\ S'\ Sl,\ X \notin dom\ S \to X \notin Sl \to S \gg S' =_{\sigma_{min}} Sl \to S'@X =_{\sigma_{min}} X$

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in vars\_of\ \ t_0$

$$H_1 \ :\ (dom\ S)\ \cap\ vars\_of\ \ P = \{\}$$
$$H_2 \ :\ (dom\ Sl) \cap\ lhvars\_of\ \ P = \{\}$$
$$H_3 \ :\ X \notin lhvars\_of\ \ P$$
$$H_4 \ :\ \forall\ s'\ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \to s' =_{\sigma_{min}} t'\langle Sl\rangle$$
$$H_5 \ :\ Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S'$$
$$s_0, t_0 \ :\ \texttt{exp}$$
$$H_6 \ :\ (s_0 =_? t_0) \in P$$

———————————————————————————————

$$s_0 =_{\sigma_{min}} t_0\langle Sl\rangle$$

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in vars\_of\ \ t_0$

$$
\begin{aligned}
&H_1 : (dom\ S)\ \cap\ vars\_of\ \ P = \{\}\\
&H_2 : (dom\ Sl) \cap\ lhvars\_of\ \ P = \{\}\\
&H_3 : X \notin lhvars\_of\ \ P\\
&H_4 : \forall\ s'\ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X,s)\}\rangle \rightarrow s' =_{\sigma_{min}} t'\langle Sl\rangle\\
&H_5 : Sl =_{\sigma_{min}} (S \gg \{(X,s)\}) \gg S'\\
&s_0, t_0 : \texttt{exp}\\
&H_6 : (s_0 =_? t_0) \in P\\
&-----------------------------------\\
&s_0 =_{\sigma_{min}} t_0\langle Sl\rangle
\end{aligned}
$$

1. We can show $\underline{s_0 =_? t_0\langle\{(X,s)\}\rangle} \in (P\backslash(s,X))\langle\{(X,s)\}\rangle$.

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in vars\_of\ t_0$

$$
\begin{array}{l}
H_1 : (dom\ S) \cap vars\_of\ P = \{\} \\
H_2 : (dom\ Sl) \cap lhvars\_of\ P = \{\} \\
H_3 : X \notin lhvars\_of\ P \\
H_4 : \forall\ s'\ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \rightarrow s' =_{\sigma_{min}} t'\langle Sl\rangle \\
H_5 : Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
s_0, t_0 : \texttt{exp} \\
H_6 : (s_0 =_? t_0) \in P \\
\text{--------------------------------------} \\
s_0 =_{\sigma_{min}} t_0\langle Sl\rangle
\end{array}
$$

1. We can show $\underline{s_0 =_? t_0\langle\{(X, s)\}\rangle} \in (P\backslash(s, X))\langle\{(X, s)\}\rangle$.
2. 1 in H_4 gives us $s_0 =_{\sigma_{min}} t_0\langle\{(X, s)\}\rangle\langle Sl\rangle$

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in vars\_of\ t_0$

$$
\begin{array}{l}
H_1 \ :\ (dom\ S)\ \cap\ vars\_of\ \ P = \{\} \\
H_2 \ :\ (dom\ Sl) \cap\ lhvars\_of\ \ P = \{\} \\
H_3 \ :\ X \notin lhvars\_of\ \ P \\
H_4 \ :\ \forall\ s'\ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \rightarrow s' =_{\sigma_{min}} t'\langle Sl\rangle \\
H_5 \ :\ Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
s_0, t_0 \ :\ \texttt{exp} \\
H_6 \ :\ (s_0 =_? t_0) \in P \\
\text{------------------------------------} \\
s_0 =_{\sigma_{min}} t_0\langle Sl\rangle
\end{array}
$$

1. We can show $\underline{s_0 =_? t_0\langle\{(X, s)\}\rangle} \in (P\backslash(s, X))\langle\{(X, s)\}\rangle$.
2. 1 in H_4 gives us $s_0 =_{\sigma_{min}} t_0\langle\{(X, s)\}\rangle\langle Sl\rangle$
3. $s_0 =_{\sigma_{min}} t_0\langle\{(X, s)\} \gg Sl\rangle$, due to substitution composition property.
4. $Y \in (vars\_of\ \ s) \rightarrow Y \notin dom\ Sl$, from H_2 and H_6

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in vars\_of\ t_0$

$H_1\ :\ (dom\ S)\ \cap\ vars\_of\ P = \{\}$
$H_2\ :\ (dom\ Sl) \cap\ lhvars\_of\ P = \{\}$
$H_3\ :\ X \notin lhvars\_of\ P$
$H_4\ :\ \forall\ s'\ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \to s' =_{\sigma_{min}} t'\langle Sl\rangle$
$H_5\ :\ Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S'$
$s_0, t_0\ :\ \texttt{exp}$
$H_6\ :\ (s_0 =_? t_0) \in P$
$---------------------------------$
$s_0 =_{\sigma_{min}} t_0\langle Sl\rangle$

1. We can show $\underline{s_0 =_? t_0\langle\{(X, s)\}\rangle} \in (P \backslash (s, X))\langle\{(X, s)\}\rangle$.
2. 1 in H_4 gives us $s_0 =_{\sigma_{min}} t_0\langle\{(X, s)\}\rangle\langle Sl\rangle$
3. $s_0 =_{\sigma_{min}} t_0\langle\{(X, s)\} \gg Sl\rangle$, due to substitution composition property.
4. $Y \in (vars\_of\ s) \to Y \notin dom\ Sl$, from H_2 and H_6
5. $Sl@X =_{\sigma_{min}} s$, same as in case 1.

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in vars\_of\ \ t_0$

$$H_1 \ :\ (dom\ S)\ \cap\ vars\_of\ \ P = \{\}$$
$$H_2 \ :\ (dom\ Sl) \cap\ lhvars\_of\ \ P = \{\}$$
$$H_3 \ :\ X \notin lhvars\_of\ \ P$$
$$H_4 \ :\ \forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \to s' =_{\sigma_{min}} t'\langle Sl\rangle$$
$$H_5 \ :\ Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S'$$
$$s_0, t_0 \ :\ \texttt{exp}$$
$$H_6 \ :\ (s_0 =_? t_0) \in P$$
$$-----------------------------------$$
$$s_0 =_{\sigma_{min}} t_0\langle Sl\rangle$$

1. We can show $\underline{s_0 =_? t_0\langle\{(X, s)\}\rangle} \in (P\backslash(s, X))\langle\{(X, s)\}\rangle$.
2. 1 in H_4 gives us $s_0 =_{\sigma_{min}} t_0\langle\{(X, s)\}\rangle\langle Sl\rangle$
3. $s_0 =_{\sigma_{min}} t_0\langle\{(X, s)\} \gg Sl\rangle$, due to substitution composition property.
4. $Y \in (vars\_of\ \ s) \to Y \notin dom\ Sl$, from H_2 and H_6
5. $Sl@X =_{\sigma_{min}} s$, same as in case 1.
6. $Sl =_{\sigma_{min}} \{(X, s)\} \gg Sl$, from 4 and 5.

Case 2.1: $s_0 =_? t_0 \neq s =_? X$, when $X \in vars\_of\ \ t_0$

$$
\begin{array}{l}
H_1 \ :\ (dom\ S)\ \cap\ vars\_of\ \ P = \{\} \\
H_2 \ :\ (dom\ Sl) \cap\ lhvars\_of\ \ P = \{\} \\
H_3 \ :\ X \notin lhvars\_of\ \ P \\
H_4 \ :\ \forall\ s'\ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X, s)\}\rangle \to s' =_{\sigma_{min}} t'\langle Sl\rangle \\
H_5 \ :\ Sl =_{\sigma_{min}} (S \gg \{(X, s)\}) \gg S' \\
s_0, t_0 \ :\ \mathtt{exp} \\
H_6 \ :\ (s_0 =_? t_0) \in P \\
\text{------------------------------------} \\
s_0 =_{\sigma_{min}} t_0\langle Sl\rangle
\end{array}
$$

1. We can show $\underline{s_0 =_? t_0\langle\{(X, s)\}\rangle} \in (P\backslash(s, X))\langle\{(X, s)\}\rangle$.
2. 1 in H_4 gives us $s_0 =_{\sigma_{min}} t_0\langle\{(X, s)\}\rangle\langle Sl\rangle$
3. $s_0 =_{\sigma_{min}} t_0\langle\{(X, s)\} \gg Sl\rangle$, due to substitution composition property.
4. $Y \in (vars\_of\ \ s) \to Y \notin dom\ Sl$, from H_2 and H_6
5. $Sl@X =_{\sigma_{min}} s$, same as in case 1.
6. $Sl =_{\sigma_{min}} \{(X, s)\} \gg Sl$, from 4 and 5.
7. $t_0\langle Sl\rangle =_{\sigma_{min}} t_0\langle\{(X, s)\} \gg Sl\rangle$

Case 2.2: $s_0 =_? t_0 \neq s =_? X$, when $X \notin vars\_of\ \ t_0$

$$
\begin{array}{l}
H_1 \ :\ (dom\ S)\ \cap\ vars\_of\ \ P = \{\} \\
H_2 \ :\ (dom\ Sl) \cap\ lhvars\_of\ \ P = \{\} \\
H_3 \ :\ X \notin lhvars\_of\ \ P \\
H_4 \ :\ \forall\ s'\ \ t', (s' =_? t') \in (P \backslash (s =_? X))\langle\{(X,s)\}\rangle \rightarrow s' =_{\sigma_{min}} t'\langle Sl\rangle \\
H_5 \ :\ Sl =_{\sigma_{min}} (S \gg \{(X,s)\}) \gg S' \\
s_0, t_0 \ :\ \mathtt{exp} \\
H_6 \ :\ (s_0 =_? t_0) \in P \\
\text{------------------------------------} \\
s_0 =_{\sigma_{min}} t_0\langle Sl\rangle
\end{array}
$$

1. Apply H_4.
2. $(s_0 =_? t_0) \in (P\backslash(s, X))\langle\{(X,s)\}\rangle$ is provable by knowing H_6, $X \notin vars\_of\ \ t_0$, and $s_0 \neq s$ and $t_0 \neq X$.